

Final project for the lecture Discrete Simulation and Reinforcement Learning

General information:

- **Registration** for the examination in the second examination period of the summer semester 2026 via the **LSF** is **necessary**
- The project can be worked on in **teams of a maximum of 2 people**, these must be registered via **e-mail** to the lecturers by **31.07.2026!**
- All team members must be able to explain all parts of the model/program code
- **Submission** of the projects by **20.09.2026**, 23:59 as a single zip file via the ILIAS form
- The **oral exams** will take place in the second examination period from **28.09. to 09.10.2026**
- In the oral exam, the project should be briefly presented using the model or code, then questions about the code of the project as well as the content of the lecture must be answered. The **presentation documents** (LaTeX/PowerPoint) **must be submitted via the ILIAS form** together with the other project documents.

Project description:

Extend the Plant Simulation model from the lecture so that it fulfills the following requirements:

- Three language variants of a board game are to be produced
- Supply materials are:
 - non-sorted, language-specific paper sheets for the instructions
 - language-neutral game material
- The instructions are to be processed (bound), this is to be represented visually
- The game material is to be processed (painted)
- One set of instructions and one set of game material each are compiled into a language- specific game
- Three distinguishable end products are created
- All end products run in unsorted order on a single conveyor belt into a storage complex, where they are sorted by a pick-and-place robot
- The pick-and-place robot can send the end products to one of two intermediate storage areas via conveyors. Alternatively, it can send them back to the entrance of the warehouse complex via a conveyor belt loop (so that it sorts them again in the further course)

- Only end products of the same type/language can be stored in the same intermediate storage areas
- If an end product of a different type is to be stored in an intermediate storage area than is already present there, all parts that are already present in the intermediate storage area are to be sent back to the entrance of the warehouse complex via a corresponding loop
- The temporary warehouses are emptied when 10 units of the same end product are present in the intermediate storage area

Based on this simulation model, a reinforcement agent is to be developed to control the pick-and- place robot in the warehouse complex. Use the methods from the lecture for this. The aim is to minimize the throughput time for 1000 manufactured games